Diffraction & Array Patterns
IN5450 Project 1

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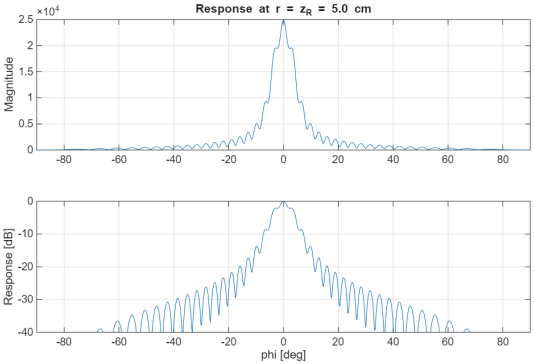
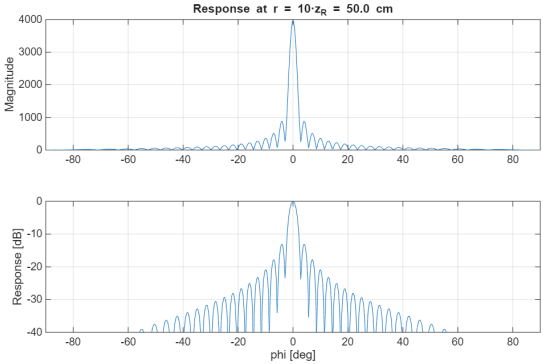
Agenda

1. Diffraction from a Line Aperture (Tasks 1-2)
2. Grating Lobes (Task 4)
3. Kaiser Window (Task 5)
4. Non-uniform Spacing (Task 6)
5. Steering (Tasks 7-8)
6. Thinned Arrays (Task 9)
7. Element Directivity (Tasks 12-13)

Task 1 - Diffraction field of a line aperture

Setup: Uniform line aperture $D = 20\lambda$, $f_0 = 3$ MHz, $c = 1500$ m/s. R-S integral evaluated via `R_S_LinAperture.m`.





Near-field ($r \approx z_R$): Irregular amplitude with ripples due to path-length interference from different parts of the aperture.

Far-field ($r \gg z_R$): Smooth sinc-like pattern - field $\propto$ spatial FT of aperture distribution.

Transition from near-field to far-field is clearly visible in the increasing smoothness of the angular response.

Code - Diffraction & beampattern.m

Core utility (used in all tasks):

```
function W = beampattern(xpos, kx, weights)
  xpos = xpos(:); weights = weights(:); kx = kx(:);
  W = exp(1j*kx.*xpos.') * weights;
end
```

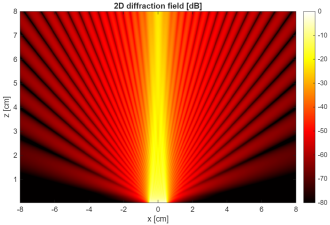
R-S diffraction setup (Tasks 1-2):

```
D = 20*lambda; dx = lambda/10;
apX = -D/2:dx:D/2;
A0 = ones(size(apX)).'; A0 = A0/sum(A0);
z_R = D^2 / (4*lambda);

% 2D field
[Resp2D, ~] = R_S_LinAperture(apX, apZ, ...
                              X, Z, lambda, A0);
```

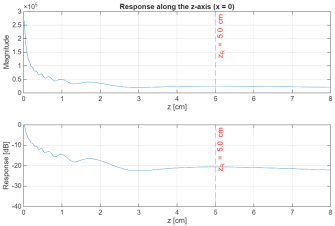
Half-circle at near-/far-field:

```
us = -pi/2:du:pi/2;
% Near-field half-circle at r = z_R
Z_near = z_R*cos(us);
X_near = z_R*sin(us);
[Resp_near, ~] = R_S_LinAperture(apX, apZ,...
                 X_near, Z_near, lambda, A0);

% Far-field half-circle at r = 10*z_R
r_far = 10*z_R;
Z_far = r_far*cos(us);
X_far = r_far*sin(us);
[Resp_far, ~] = R_S_LinAperture(apX, apZ,...
                 X_far, Z_far, lambda, A0);
```

Task 2 - 2D field & near-/far-field boundary



2D field: beam confined near aperture, spreading at larger z.



On-axis: amplitude oscillates in NF, decays as $1/\sqrt{z}$ beyond z_R.

Derivation:

$$\Delta\phi_{\mathbf{max}} = \frac{\pi D^2}{4\lambda z}$$

Setting $\Delta\phi_{\mathbf{max}} = \pi/2$ (path diff. $= \lambda/4$):

$$\boxed{z_R = \frac{D^2}{4\lambda}} = 100\lambda = 5\,\text{cm}$$

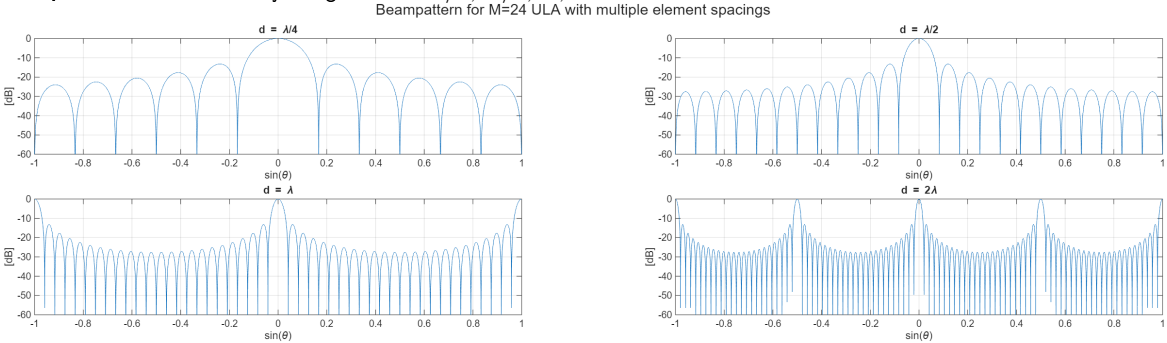
Differs from piston formula $2D^2/\lambda$ which uses a stricter phase criterion for 2D apertures.

At $r = z_R$: NF artifacts; sidelobes not stabilised.
At $r = 10z_R$: Smooth sinc, first SL $= -13.2$ dB.

Ref: S. Holm, *IN5450 Diffraction: from waves to signal processing*, UiO, 2019.

Task 4 - Effect of element spacing on grating lobes

Setup: $M = 24$ ULA, unity weights, $d = \lambda/4,\ \lambda/2,\ \lambda,\ 2\lambda$.



- $d = \lambda/4$: Wide main lobe (8.5°), no grating lobes; small aperture gives poor resolution.
- $d = \lambda/2$: BW $\approx 4.2^\circ$, max SL $= -13.2\,\mathrm{dB}$. No grating lobes gives good balance.
- $d = \lambda$: Grating lobe at $u = \pm 1$ ($\theta = \pm 90^\circ$). Spatial Nyquist violated.
- $d = 2\lambda$: Multiple grating lobes at $\Delta u = \lambda/d = 0.5$. Ambiguous direction finding.

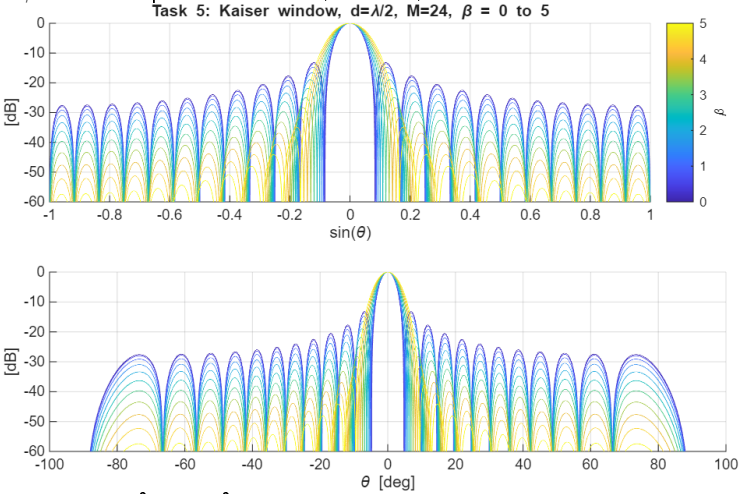
Code - Grating lobe analysis (Task 4)

```
M = 24; weights = ones(M,1)/M;
spacings = [1/4, 1/2, 1, 2]; % in units of lambda
u = linspace(-1, 1, 10000); kx = 2*pi/lambda * u;

for i = 1:4
    d    = spacings(i) * lambda;
    xpos = (0:M-1).' * d;        % ULA positions
    W    = beampattern(xpos, kx, weights);

    Result = analyzeBP(u, W);
end
```

analyzeBP returns beamwidths (at -3 dB and -6 dB) and maximum sidelobe level. Grating lobes appear when $d \geq \lambda$ since the beampattern is periodic with $\Delta u = \lambda/d$.

Task 5 - Kaiser window on $M = 24$, $d = \lambda/2$

Setup: Kaiser window $\beta = 0$-5. Report beamwidths, max SL, WNG.



WNG $= |\sum w_m|^2 / \sum |w_m|^2$. Uniform weighting maximises WNG at $10 \log_{10}(24) = 13.8$ dB.

Task 5 - Results

β	-3 dB BW (°)	-6 dB BW (°)	Max SL (dB)	WNG (dB)
0.0	4.21	5.74	-13.2	13.8
1.0	4.37	5.97	-14.7	13.8
2.0	4.78	6.57	-19.0	13.6
3.0	5.33	7.35	-25.0	13.2
4.0	5.88	8.18	-31.7	12.7
5.0	6.43	8.96	-38.0	12.3

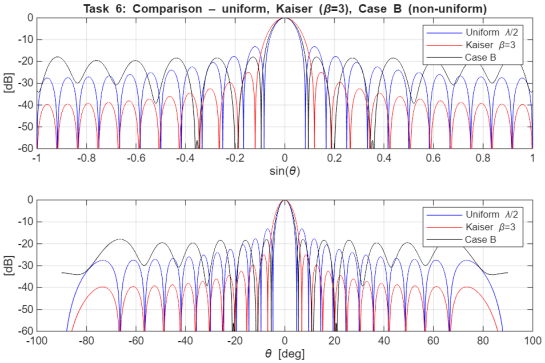
- **Advantage**: SL drops from -13 to -38 dB - critical for rejecting interferers.
- **Disadvantage**: Main lobe broadens ($4.2^\circ \rightarrow 6.4^\circ$); WNG drops slightly ($13.8 \rightarrow 12.3$ dB).
- Each $\sim$12 dB of SL suppression costs $\sim 1^\circ$ of mainlobe broadening.

Code - Kaiser window & WNG (Task 5)

```
d = lambda/2; xpos = (0:M-1).' * d;
betas = 0:0.5:5;

for i = 1:length(betas)
    w = kaiser(M, betas(i)); % Kaiser window
    w = w / sum(w);          % normalise so sum(w) = 1

    W = beampattern(xpos, kx, w);

    % White noise gain: WNG = |sum(w)|^2 / sum(|w|^2)
    % Since sum(w) = 1: WNG = 1 / sum(w.^2)
    WNG = 1 / sum(w.^2);

    Result = analyzeBP(u, W);
end
```

Higher β concentrates weight energy in fewer elements $\Rightarrow$ SL drops but BW broadens. WNG penalises non-uniform weights: $\mathrm{WNG} = |\sum w_m|^2 / \sum |w_m|^2$.

Task 6 - Uniform vs. Kaiser vs. Case B



Array	WNG	−3 dB	Max SL
Uniform	13.8	4.21°	−13.2
Kaiser $\beta = 3$	13.2	5.33°	−25.0
Case B	13.8	4.55°	−17.9

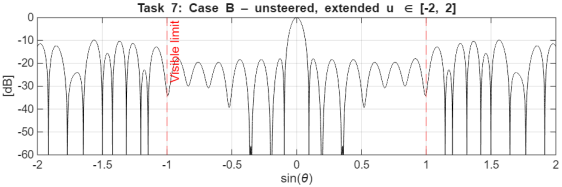
Case B uses uniform weights and achieves full WNG, but element positions are optimised to reduce SL. BW is slightly wider than uniform but much narrower than Kaiser.

≈ 5 dB SL improvement over uniform with only 0.3° BW cost.

Optimising element positions reduces SL without sacrificing noise robustness.

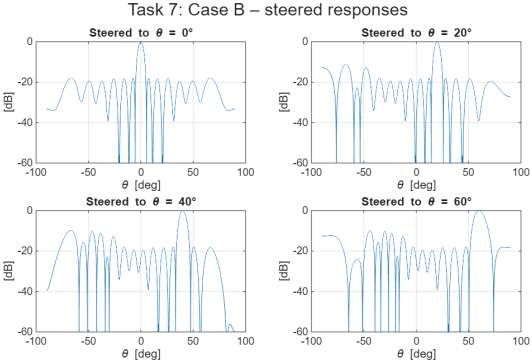
Task 7 - Steering the Case B array

Unsteered, $u \in [-2, 2]$:



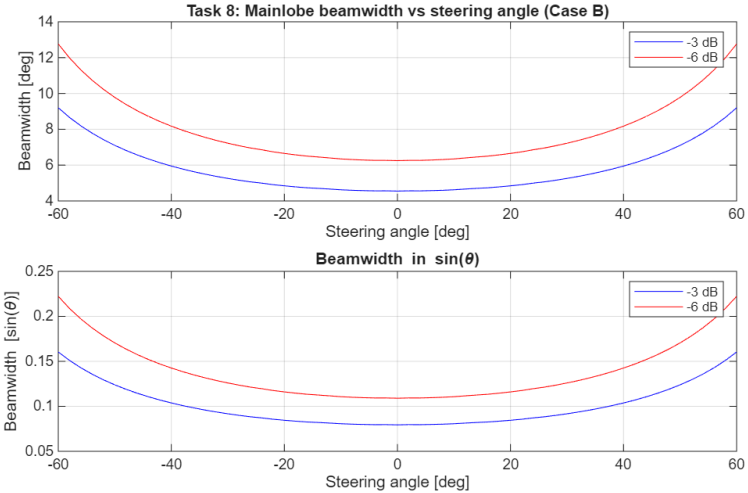
Outside the visible region the beampattern continues. When steered, this structure can enter the visible region.

Steered ($\theta_s = 0°, 20°, 40°, 60°$):



Main lobe broadens and SLs grow at large steering angles.

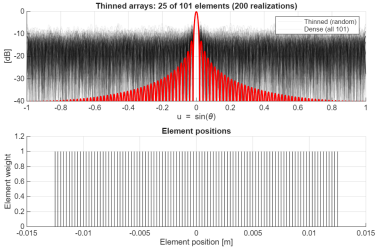
Task 8 - Beamwidth vs. steering angle (Case B)



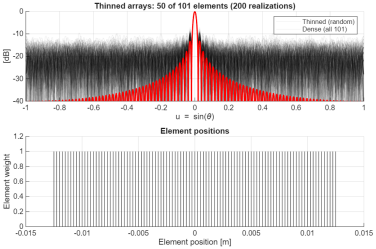
Result: BW increases proportionally to steering angle magnitude.

Task 9 - Random thinning (200 realisations)

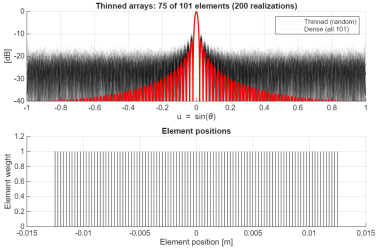
Setup: $M = 101$, $d = \lambda/2$. Randomly select N elements (outer two fixed).



$N = 25$



$N = 50$



$N = 75$

Red = dense $N = 101$ reference. Grey spread narrows as N increases; SL floor converges to dense array.

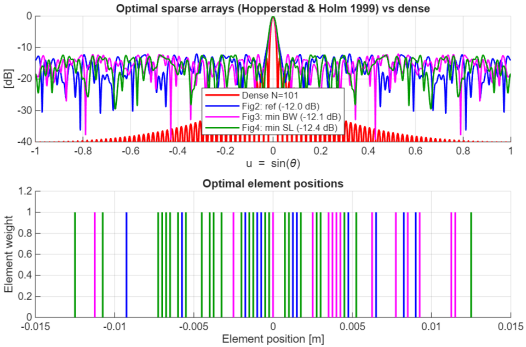
Task 9 - Thinning statistics & optimal arrays

Random thinning:

N	BW (°)	$\sigma_{\mathbf{BW}}$	Max SL	Mean SL
25	0.96	0.07	−7.6	−17.4
50	0.98	0.04	−11.6	−22.0
75	0.99	0.03	−13.0	−26.4
101	1.00	-	−13.3	−40.7

H&H optimal ($N = 25$):

Design	BW	Max SL	Mean SL
Reference	1.41°	−12.0	−16.9
Min BW	1.18°	−12.1	−16.2
Min SL	1.43°	−12.4	−16.8



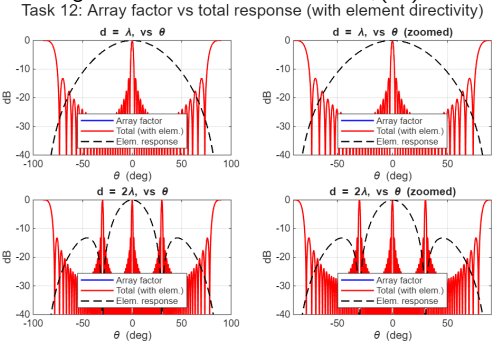
Code - Random thinning (Task 9)

```
NEls = 101; d = lambda/2; NRealizations = 200;
xpos_dense = (-(NEls-1)/2 : (NEls-1)/2).' * d;

for ni = 1:length([25, 50, 75])
    NPos = NPos_list(ni);
    BW3_all = zeros(NRealizations,1);
    maxSL_all = zeros(NRealizations,1);

    for r = 1:NRealizations
        % Random subset, keep outer two elements fixed
        pos = unique(ceil((NEls-2)*rand(1,NPos*2)), 'stable');
        ElPos = [-(NEls-1)/2, sort(pos(1:NPos-2))-(NEls-1)/2,...
                 (NEls-1)/2] * d;

        w = ones(NPos,1) / NPos;
        W = beampattern(ElPos.', kx, w);

        R = analyzeBP(u, W);
        BW3_all(r) = rad2deg(R.Three_dB);
        maxSL_all(r) = R.maxSL;
    end
end
```

Fixed outer elements preserve aperture $\Rightarrow$ BW unchanged. Statistics over 200 trials characterise the SL floor.

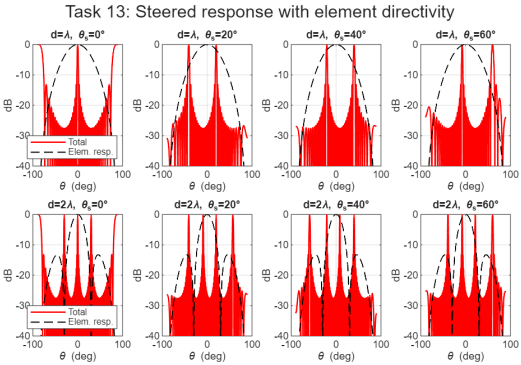
Task 12 - Including element directivity

Setup: $M = 24$, $d = \lambda$ and 2λ. Rectangular element of width d: $H_e(k_x) = \mathrm{sinc}(k_x d/2\pi)$.



- $d = \lambda$: Element response zero at $u = \pm 1$ coincides with grating lobe $\Rightarrow$ suppressed at broadside. Element envelope spans full visible region.
- $d = 2\lambda$: Element zeros at $u = \pm 0.5, \pm 1$ coincide with grating lobes $\Rightarrow$ also suppressed at broadside.

Task 13 - Steering with element directivity



Steering shifts the array factor in u-space, but H_e is fixed since elements dont move.

- $d = \lambda$: Grating lobes shift away from element zeros and reappear. At $\theta_s = 60^\circ$ a GL is partially visible but still attenuated.
- $d = 2\lambda$: At $\theta_s = 20^\circ$ the first GL near broadside exceeds the steered main lobe - element pattern works *against* the beam. Usable scan $\lesssim 10^\circ$.

Conclusion: Element directivity provides excellent GL suppression at broadside but diminishes with steering. For $d > \lambda/2$ the scan range is fundamentally limited.

Code - Element directivity (Tasks 12-13)

```
% Element response for rectangular element of width d:
% H_e(kx) = sinc(kx*d/(2*pi))
elem_resp = @(d) sinc(kx * d / (2*pi));

for di = 1:2
    d = d_list(di); % lambda, 2*lambda
    xpos = (-(M-1)/2 : (M-1)/2).' * d;
    w = ones(M,1) / M;

    W_array = beampattern(xpos, kx, w); % array factor
    He      = elem_resp(d);             % element pattern
    W_total = W_array .* He;            % total response
end

% Steering: shift kx by steering direction
for si = 1:length(steer_angles)
    theta_s = deg2rad(steer_angles(si));
    kx_steer = 2*pi/lambda * (u - sin(theta_s));
    W_steered = beampattern(xpos, kx_steer, w) .* He;
end
```

Total response = array factor $\times$ element pattern. Steering shifts $W(k_x)$ but $H_e(k_x)$ stays fixed, so grating lobes reappear when steered away from broadside.

Summary

1. **Diffraction (1-2)**: R-S integral reproduces near-field to far-field transition; $z_R = D^2/(4\lambda)$ for line aperture.
2. **Grating lobes (4)**: Appear for $d \geq \lambda$; $d = \lambda/2$ is the aliasing-free limit.
3. **Kaiser window (5)**: $\sim$12 dB SL suppression per 1° BW broadening; modest WNG cost.
4. **Non-uniform spacing (6)**: Case B achieves -17.9 dB SL with unity weights and full WNG.
5. **Steering (7-8)**: BW increases with greater magnitude steering angle.
6. **Thinning (9)**: Preserves BW but raises SL floor; optimised placement $\approx$5 dB better than random.
7. **Element directivity (12-13)**: Element response suppresses grating lobes at broadside; steering breaks the alignment.